Hugo Marquerie 13/01/2026

Ejem continuidad derivadas laterales imp dini

Ejemplo 1. Sea $f \in \mathcal{L}^1\left(\left[-1/2, 1/2\right)\right)$ 1 periódica, continua en $x_0 \in \left(-1/2, 1/2\right)$ y tal que existen $f'(x_0^-)$ y $f'(x_0^+)$. Entonces, f satisface el criterio de Dini en x_0:

$$\begin{aligned}\int_{|t|<\delta} \left|\frac{f(x_0+t)-f(x_0)}{t}\right| \mathrm{d}t &= \int_{-\delta}^{0} \frac{|f(x_0+t)-f(x_0)|}{-t}\,\mathrm{d}t + \int_0^\delta \frac{|f(x_0+t)-f(x_0)|}{t}\,\mathrm{d}t \\ &= \int_0^\delta \frac{|f(x_0-s)-f(x_0)|}{s}\,\mathrm{d}s + \int_0^\delta \frac{|f(x_0+t)-f(x_0)|}{t}\,\mathrm{d}t.\end{aligned}$$

Como $f'(x_0^+)$ existe, $\exists \delta_1 > 0$ tal que $\forall t \in (0, \delta_1) : \left|\frac{|f(x_0+t)-f(x_0)|}{t} - f'(x_0^+)\right| < 1$ y como $f'(x_0^-)$ existe, $\exists \delta_2 > 0$ tal que $\forall s \in (0, \delta_2) : \left|\frac{|f(x_0-s)-f(x_0)|}{s} - f'(x_0^-)\right| < 1$. Entonces, tomando $\delta = \min\{\delta_1, \delta_2\}$, tenemos que

$$\int_{|t|<\delta} \left|\frac{f(x_0+t)-f(x_0)}{t}\right| \mathrm{d}t \leqslant \int_0^\delta \left(1 + \left|f'(x_0^-)\right|\right) \mathrm{d}s + \int_0^\delta \left(1 + \left|f'(x_0^+)\right|\right) \mathrm{d}t < \infty.$$

Referenciado en

- `prop-criterio-dirichlet`

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